

Efficiency & market failure

Economic efficiency requires avoiding waste of resources. When labor is unemployed and factories lay idle (as occurs in serious recessions), their potential output is lost. If these resources were employed, total output would be increased and hence everyone could be made better off. However, the full employment of resources is not by itself enough to prevent the waste of resources. Even when resources are being fully used, they may be used inefficiently.

Productive efficiency: Productive efficiency refers to the efficient production of any bundle of products that is being produced. It occurs when it is impossible to reallocate resources so as to produce more of some product without producing less of some other product.

Productive efficiency has two aspects, one concerning the allocation of resources within each firm, and one concerning the allocation of resources among the firms in an industry. The first condition for productive efficiency is that each firm should produce any given output at the lowest possible cost. In the short run, with only one variable factor, the firm has no problem of choice of technique: it merely uses enough of the variable factor to produce the desired level of output.

In the long run however more than one method of production is available. Productive efficiency requires that the firm use the least costly of the available methods of producing any given output. This means that firms will be located on, rather than above, their long run average cost curves.

Productive efficiency requires that each firm produce its given output by combining factors of production in such a way that the ratio of the marginal products of each pair of factors is made equal to the ratio of their prices.

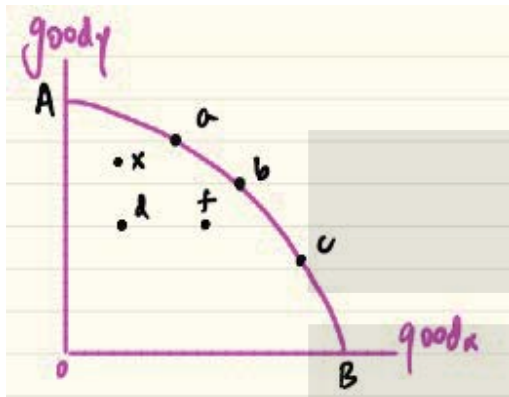
This is the same thing as saying that \$1 spent on every factor should yield the same output. If this is not so, the firm can reduce the resource costs of producing its given output by altering the inputs it uses. It should substitute the input for which \$1 of expenditure yields the higher output for the input for which \$1 of expenditure yields the lower output.

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There is a second condition for productive efficiency that ensures that the total output of each industry is allocated among its individual firms in such a way that the total cost of producing the industry's output is minimized.

Productive efficiency requires that the marginal cost of producing its last unit of output must be the same for each firm in any industry.



All the points on the production possibility curve show that they are productively efficient but not all points are allocative efficient. The curve shows all the combination of two goods that can be produce when the economy's resources are fully employed and being used with productively efficiency.

Allocative efficiency: Allocative efficiency concerns the allocation of resources among products; in other words. Allocative efficiency concerns the choice among alternative points on the production possibility curve.

Allocative efficiency is defined as a situation in which it is impossible to change the allocation of resources in such a way as to make some person better off without making some other person worse off. Changing the allocation of resources implies producing more of some goods and less of others, which in turn means moving from one point on the production possibility curve to another. This is called changing the mix of production. The economy's allocation of resources is efficient when, for each good produced, its marginal cost of production is equal its price.

Allocative efficiency occurs when firms produce the combination of goods and services that are most wanted by consumers. Allocative efficiency gives consumers maximum satisfaction at their current level of income. The implication is that

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there is no waste and both producers and consumers are satisfied with what is produced. The price that they are willing to pay reflects their preferences and the benefits they derive from consumption. Allocative efficiency represents the additional benefit derived from the consumption of one more unit of a particular good. The marginal cost, the cost of producing one more unit of a good, is a measure of the opportunity cost of the resources used to produce this last unit. So, where the price equals the marginal cost, consumers are prepared to pay what it costs to produce the last unit. It is only where this occurs that resources are efficiently allocated.

Pareto optimality: One particular type of optimality is associated with the Italian economist Vilfredo Pareto (1848-1923). Pareto optimality (or Pareto efficiency) refers to a situation where it is not possible to reallocate scarce resources to make someone better off without making someone else worse off. For this situation to be achieved, there needs to be both productive and allocative efficiency. This optimal allocation of resources means that inputs into the production process are used in the most efficient way (productive efficiency) and that the output produced yields the maximum possible utility or satisfaction to consumers (allocative efficiency).

Pareto first put forward this condition in 1909. If an allocation of resources is Pareto inefficient, this means that it would be possible to make at least one person better off without making anybody worse off

Dynamic efficiency results from improvements in technical or productive efficiency over a period of time. For example, new products may appear in markets as a result of research and development, invention and innovation. It is important to understand the difference between invention and innovation; innovation is the practical application of inventions to general products that can be sold in the market place. Innovation is the hardest part of technical progress, i.e. transforming a good idea into something that is successful in that it is workable and profitable.

New methods of production could also be introduced, incorporating the latest technology in order to increase productivity. There could also be the introduction

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of radically different methods of management, resulting from investment in human capital.

Both productive and allocative efficiency can be regarded as examples of static efficiency, i.e. they are concerned with the allocation of scarce resources at a particular moment in time. Dynamic efficiency, in contrast, is concerned with the allocation of scarce resources over a period of time.

GOVERNMENT INTERVENTION IN THE PRICE SYSTEM

Market failure occurs when freely functioning markets fail to deliver an efficient allocation of resources. The result is a loss of economic and social welfare. Market failure exists when the competitive outcome of markets is not efficient from the point of view of society as a whole. This is usually because the benefits that the free market confers on individuals or businesses carrying out a particular activity diverge from the benefits to society as a whole.

PRIVATE COSTS are the costs which are incurred by the producers when they produce goods and services e.g. cost of labor, raw materials, and machines and so on. Consumers also incur private costs when they buy different goods and services. The greater the production or consumption of goods and services, the higher will be the private costs to producers and consumers. These are also called internal costs in the sense that they are borne by those who produce or consume goods and services.

PRIVATE BENEFITS are the benefits which are received by the producers when they sell goods and services which they produce. The revenue which the producer receives is the private benefit. Consumers also receive the benefits when they consume the goods and services. The utility or satisfaction which they receive is the private benefit. These benefits are also called internal benefits in the sense that they are received only by those who produce or consume goods and services.

EXTERNALITIES are third party effects arising from the production and consumption of goods and services for which no appropriate compensation is paid. Externalities cause market failure if the price mechanism does not take account of the external cost and benefits of production and consumption. Externalities occur

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outside of the market i.e. they affect economic agents not directly involved in the production or consumption of a particular goods or service. Since externalities do not go through the market, they do not have prices attached to them.

NEGATIVE EXTERNALITIES occur when production or consumption impose external costs on third parties outside of the market for which no appropriate compensation is paid.

For example consider the situations in which a business dumps the waste products from its production process in to a nearby river or in which an individual litters a park or beach. Obviously, a cost is involved in these actions. When the firms pollute the water, people downstream suffer consequences. They may not want to swim in or drink polluted water. They may also be unable to catch fish as before because of pollution. In the case of littering, the people who come along after our litter has cluttered the park or the beach are the ones who bear the costs. The costs of these actions is borne by people other than those who commit these actions, the creator of the costs is not the sole bearer. The costs are not internalized by the individual or firm; they are external.

Some examples are:

1. Smokers ignore the unintended but harmful impact of passive smoking on non-smokers or tribute to the growing costs of treating smoking-related diseases.
2. Acid rain from power stations
3. Air pollution from the road use and traffic congestion the social costs of drugs and alcohol abuse
4. The environmental damage cause by the intensive use of fertilizers in agriculture

POSITIVE EXTERNALITIES exist when third parties benefit from the spillover effects of reproduction and consumption. Examples are:

1. Social returns from increase investment in education and training. If one person gets educated, it benefits the entire society. The person can work as a trained worker, or he/she can be a trained professional worker at any job.

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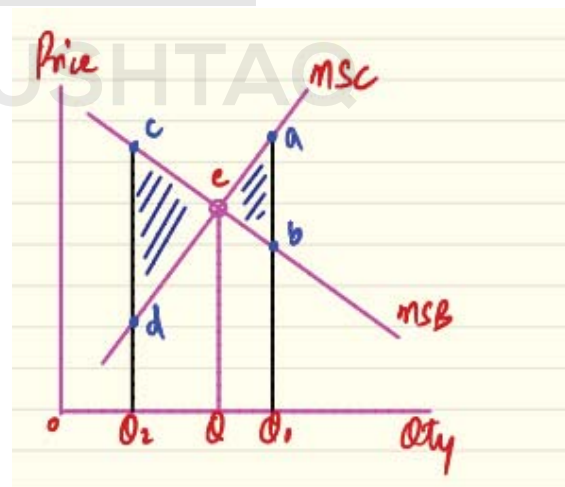
2. Positive benefits from improvements in the quality of health care and medical research.
3. Improved health outcomes arising from vaccination program. If one gets vaccinated, it prevents others from contracting the disease from that person, thus a third party is not involved in the transactions.
4. Provision of flood protection system and installation of more effective fire safety equipment
5. The social benefits to be derived from restored historic buildings and monuments.
6. External benefits from people usage of public libraries and museums.

SOCIAL COSTS: It is the total cost which a society bears as a result of production and consumption activity in society. It is the sum of private costs (costs to the producers and consumers of goods and services) and external costs (costs which the third parties bear as a result of production and consumption of others).

$SC = PC + EC$. If social cost exceeds private cost that means there will be external cost/negative externalities

SOCIAL BENEFITS: These are the total benefits which a society receives as a result of production and consumption activities in the society. It is the sum of private benefits (benefits received by the producers and consumers) and external benefits (benefits received by the third parties). $SB = PB + EC$. Net social benefit refers to social benefit minus social cost. Net external benefit refers to external benefit minus external cost.

SOCIALLY OPTIMUM LEVEL OF OUTPUT: Society achieves the maximum social welfare at a point where marginal social cost is equal to marginal social benefit. At this point the total benefits of the society are maximized and the cost of producing the last unit is equal to the value which the society places to that unit.



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If the society produces less than socially optimum level of output the total welfare of the society will not maximize. If $MSB > MSC$, the benefit of the last unit produced will exceed the cost of producing that unit. In this situation the society can achieve more benefits if the output increases until all the excess of benefits over and above the cost are achieved. On the other hand if the output produced is more than the socially optimum level, the society faces higher costs on all the units i.e. $MSC > MSB$. In this case the society will gain more benefits if the output is reduced until all the excess of costs over the benefits are eliminated and the society moves to the socially optimum output.

CAUSES OF MARKET FAILURE AND GOVERNMENT INTERVENTION: There are several important circumstances under which markets fail to allocate resources with reasonable efficiency

1. **PUBLIC GOODS:** Goods that are neither excludable nor rivalrous are called public goods or sometimes called collective consumption goods. The classic case is national defense. An army of a given size protects all the nation's citizens equally no matter how many citizens are there and whether or not a particular individual pays taxes to support it. Other examples include lighthouses, street lights (it is a public good because no one can be prevented from enjoying the benefits), radio broadcasts etc. In case of public goods the benefits it confers on consumers can be extended to others at zero cost.

Because public goods are non-excludable, private firms will not provide them. The main reason for private sector not to produce public goods is that the price charged must be zero. Consumption of public goods is non-rival, meaning that the consumption of a public good or service by one individual does not preclude consumption by another individual; it does not diminishes the quantity for other individual. Secondly, consumption is non-excludable. This means that consumption by one individual makes it impossible to exclude any other individual from having the opportunity to consume. Of course, in this situation, private sector businesses are unlikely to consider providing pure public goods because there is no mechanism to charge the price as many consumers can take a free ride on such goods because of non-excludability.

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The provision of pure public goods is therefore a cause of market failure. Since market economy does not provide public goods, a government rectifies the market failure by providing the entire output. The production of public goods has to finance by the government as people are able to consume public goods without paying for them.

Another characteristic of public good is non rejectable, an individual cannot reject public good. For example, one day an individual decides that he/she is not requires the services of defense, this action does not means that defense services will now protect the whole country expect that individual.

2. **LACK OF INFORMATION:** Consumers and producers require complete information if they are to make efficient choices. In the market economy we assume that all agents in the market have perfect information about the availability of goods and services and also the prices charged by suppliers. Consumers can make purchasing decisions on the basis of full and free information on the products that they are buying. In reality all of experience information deficits which can lead to misallocation of resources. Information failure occurs when people have inaccurate, incomplete, uncertain or misunderstood data to make potentially wrong choices. Consumers might under/overestimate the private benefit from consuming a particular good or service. The classic case of this is the demand for health or education - where consumers may underestimate the long term benefit from investing in extra education or buying a specific form of health treatment.

Asymmetric Information

It was explained that the under consumption of merit goods and the over consumption of demerit goods is because consumers make choices based on imperfect information. It was assumed that all consumers have the same information when making decisions on what or what not to consume. This is not always true. The quality of information available to consumers varies and can lead to a misallocation of resources.

In many markets in real life, sellers generally know more about what is being sold than buyers. A typical example is the market for used cars or scooters. Buyers

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find it difficult to know whether they are buying a good quality car or scooter, or one of low quality. Because they do not know, buyers will only be prepared to offer an average price for what may turn out to be a good quality car or scooter. The owners of good quality cars or scooters would not want to sell at an average price. The result is that the only cars or scooters sold would be poor quality. The market has failed; poor quality cars and scooters have driven good quality ones out of the market.

This is an example of asymmetric information. Asymmetric information occurs when one party in the market, usually the seller, has some information that the other party, usually the buyer, does not have. This situation applies in the market for used cars and scooters. Buyers are often more concerned about the color of the car, for example, and not whether it has been repaired following an accident. Another example is healthcare. Medical practitioners know far more about medical conditions than their patients and are in a better position to be able to advise what treatment should be given to their patients.

In other markets, the buyer may have some information that is unknown to the seller. Suppose you want a bank loan to set up a business. The bank has to take a risk by assuming that the information provided by you as the borrower is correct. After all, only the borrower knows whether they will be able to pay back what is owed. A similar situation applies in the insurance market where the seller has usually to accept the health details provided by the buyer. Some important details may not be provided.

Adverse selection and moral hazard

There are two types of asymmetric information:

- **Hidden characteristics**, when only one party knows more about a situation than the other party. An outcome of this situation is adverse selection. For example, Buyers of used cars may end up purchasing more low quality cars in the market, reducing the market for good quality vehicles. As another example, unhealthy people could purchase more life insurance than they need while healthy people may not purchase any life insurance.

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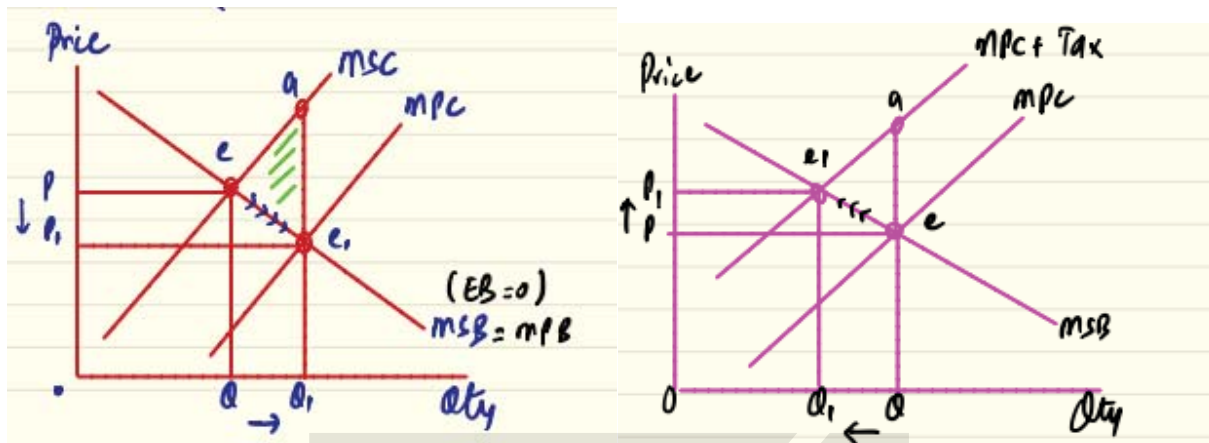
- **Hidden actions**, when one party takes actions that the other party cannot observe but which affect both of them. This may apply in the case of a bank loan where the person who has obtained the loan takes a bigger risk in setting up a business. With insurance, a policy holder may attempt to do something that is prohibited by the terms of the policy. These situations involving hidden actions are known as moral hazard.
3. **EXTERNALITIES**: Socially optimum level of output is where $MSC = MSB$. But in the market economy private producers are concerned with the private costs and private benefits only i.e. they aim to maximize profits. They do not take external costs and benefits into account. When production involves external costs, the output is more than socially optimum level i.e. there is over allocation of resources. On the other hand when the production involves external benefits, the output produced is less than socially optimum level i.e. there is under allocation of resources. Both these situations lead to market failure.

Demerit goods: These are the goods which are harmful for the society and have immense external costs. Examples include cigarettes, alcohol and drugs. These are over produced and undercharged by the market economy because people are unaware of the true cost of consuming them. They are overproduced which results in an increase in supply therefore a fall in the price level, a fall in price level therefore increases quantity demanded by individuals, so there is too much production and consumption of these goods.

Cigarettes cause unpleasant smoke which is dangerous to people in the area who are forced to become passive smokers. They inflate the national health bill because both the smokers and the passive smokers get sick. But smokers will not stop, perhaps are unable to do so because they are addicted, they have the lack of information or knowledge regarding the effects of these goods.

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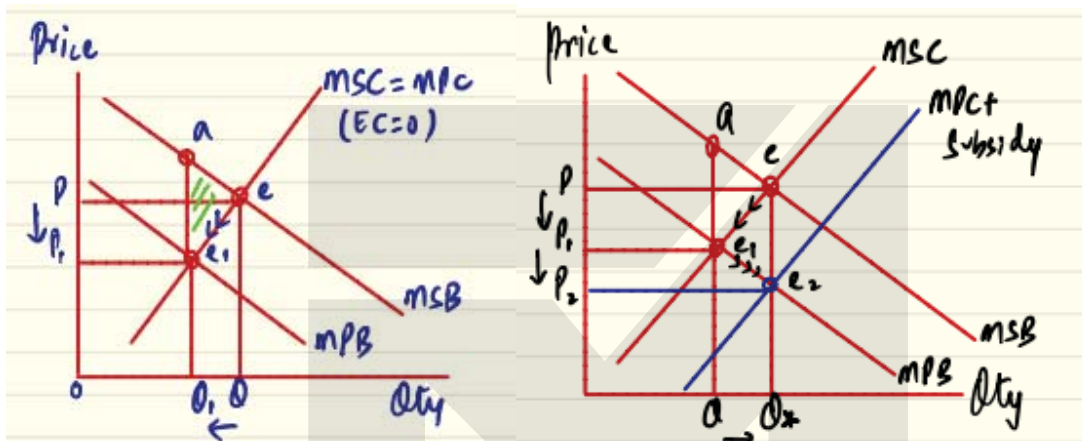
Too many demerit goods are demanded, so the government steps in and taxes highly in order to reduce the consumption and to raise the revenue which is needed anyway to spend treating the sick ones. The government advertises heavily to try to persuade people stop smoking and the young not to start and is seriously considering banning smoking in all public work places. The government imposes indirect tax on the producer causing the MPC curve to shift to left to become MSC curve. The vertical distance between the MSC and MPC curves is the per unit tax, the price of the good rises and the output fall to the socially optimum level. The burden of indirect tax on producers and consumers will depend on the elasticity of demand and supply. These goods have inelastic demand, so taxing these goods won't be effective as consumers will be ready to pay higher price, only benefit will be to the government in this situation as they will receive tax revenue. It is important to note that measuring per unit tax is also difficult for the government i.e. how much to tax these goods, for that they have to do research which can increase their administrative cost and every decision of the government will involve opportunity cost. If the per unit tax is set too high, this might bring inflationary pressure and on the other hand, if too low then this will be ineffective.

Merit goods: There is a class of goods whose consumption benefits are higher for the society as compared to individuals, these goods are called merit goods e.g. education and health. When a person gets educated, the acquired knowledge and skills increase the income of individual. To society actual benefits are higher productivity, new and better products, civilized people etc. in such a case a market economy under allocates scarce resources to the provision of merit goods. These

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goods, due to lack of information/knowledge are under consumed i.e. less demanded, which results in fall in price and provides signal to producers to under allocate the resources i.e. too little consumption and production. When talking about merit goods a government believes that consumers derive greater benefit from a good than consumers themselves realize. E.g. Education is considered to be merit good as it benefits both the recipient and society as a whole.



Examples include:

- Labor training in firms; one firm may do little, as it knows that when a trained worker leaves, someone else benefits - but the first firm paid for all the training!
- Education generally
- Governments often subsidize the cost of entrance to museums because social benefits exceed private benefit.
- The growing of flowers in a private garden results in positive externalities as social benefits exceed private benefits.
- Health generally, especially in poor third world countries.
- The provision of playing field at or near schools so that the health and sporting skills of the children improves.
- Free museums and art gallery that can encourage the poor and uneducated to widen their horizons educate themselves and generally improve.

Demand for education is equal to marginal private benefits. If benefits are less than costs, a private citizen will not acquire education. In the market economy the

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amount of education supplied is where demand intersects with supply curve. Since education has positive externalities or where social benefits exceed private benefits, the market economy under allocated resources for education.

Government provides subsidies to the activities that lead to positive externalities. This reduces the costs of production for suppliers and encourages a higher output. For example the government may subsidize state health care; public transport or investment in new technology for schools and colleges to help spread knowledge and understanding. There is also a case for subsidies to encourage higher levels of training as a means to raise labor productivity and improve our international competitiveness.

The reasons for market failure

Market failure	Explanation
Public good	These are goods which would not be provided in a market economy because it would be impossible to charge price for them because of the "free rider" problem.
Merit goods	These are goods or services which are regarded as being socially desirable and which would be under consumed and under produced in a market economy.
Demerit goods	These are goods which are regarded as being socially undesirable and which would be over consumed and over produced in a market economy.
Information failure	This is where there is a lack of full information and so the allocation of resources is less efficient than would otherwise be. This information failure is a major reason why consumers make the wrong decisions in relation to the consumption of merit and demerit good.

Government failure A government can attempt to intervene in a market economy to try to overcome the existence of a market failure, but there is a possibility that such intervention

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might create further distortions in the market.

Externalities

Externalities are costs or benefits which can affect third parties and are sometimes referred to as "spill-over" effects. There can be both positive and negative externalities and these can be related to both production and consumption.

Imperfect competition

There are different forms of imperfect competition that can exist in an economy and one of these would be the existence of a monopoly where one firm controls a market. If a firm has monopoly power in a market, it will operate very differently from a firm in perfect competition. For example, the price is likely to be higher and the output lower than would be the case in perfect competition. Also, abnormal or supernormal profits can exist in the long run as well as the short run.

Inequality in the distribution of income and wealth

A market economy may lead to a very unequal distribution of income and wealth, and this will have the effect of giving some people more influence than others in the economy.

Price instability

Price acts as a signal in a market economy and it has sometimes been described as an 'invisible hand', responsible for the allocation and reallocation of resources in an economy. However, in some situations, there can be a great deal of price instability in a market economy where prices can change quite rapidly in a relatively short period of time. This can, for example, be a feature of agricultural markets.

Factor immobility

The factors of production should be able to move easily for one sector to another in a market economy, but this is not always the case because of the existence of geographical and occupational immobility of labor.

Asymmetric

Not all parties will have the same knowledge and one party

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information Moral hazard could exploit this advantage.

Moral hazard A person or organization may be more likely to take a risk in the knowledge that someone else will bear the cost of that risk.

Government policies to correct negative and positive externalities

Regulations and the work of regulatory bodies is another way to overcome market failures caused by negative production externalities. Consider the case of a mining company that pollutes the surrounding countryside. The government might intervene by setting standards that restrict the amount of polluted waste that can be legally dumped. The government would then need to regulate and inspect the company to make sure that these restrictions are enforced. If regulations are being abused, a regulatory body can impose large fines on the company. If the damage persists, the firm could be forced to close.

Exhaust fumes from vehicles pollute the atmosphere and, to reduce this problem, governments set legal limits on the amount of carbon particles that are emitted from a vehicle's exhaust system. Virtually all countries have this type of regulation in place. Collectively, this type of regulation is sometimes referred to as a 'command and control' approach.

Deregulation involves the removal of any regulations that act as barriers to entry for new firms seeking to enter a market. The purpose of deregulation is to encourage competition between new firms and those that are established in an industry. New firms are likely to be more efficient; prices will fall and the polluting firms are likely to be forced to exit the industry as they are unable to compete.

Property rights: This is a further way in which a market-based solution can be used to correct negative production externalities. It involves property rights, the term used to indicate how owners can use their assets. If a property right can be established, a market solution can usually be reached through agreement to extend these property rights to those causing the negative externality. In many cases, property rights cannot be established, such as in cases involving the air and sea.

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The result is that either no action can be taken or government intervention corrects the market failure.

The absence of property rights leads to negative externalities of production. By changing or extending property rights, owners of property can prevent others from imposing costs on their property or they could charge them if they do impose costs.

Two possibilities can occur:

- If a polluting firm has property rights, then those who are affected by its activities could pay the polluter to reduce the scale of activity and therefore the pollution. In principle the polluter would require payment equal to the loss of profit. An example of this might be a situation where a long-established but polluting firm is asked to relocate as its emissions are affecting people who live in the surrounding area. The firm has property rights, so why should it move? If it does move, the firm could argue that it should not have to pay the cost of a move as it could stay where it has always been. The government could threaten to use new regulations, so bargaining is likely to take place until a solution is reached.
- If those affected by negative externalities are assigned property rights, then a polluting firm may be sued and obliged to compensate them in some way. In principle the charge would be equal to the marginal external cost. There is likely to be a difference in the bargaining power of the two parties. An individual with property rights is likely to experience problems if taking on a powerful multinational company. The chance of success will be greater if others with property rights join the action against the polluting firm.

Pollution permits: A fourth form of intervention is through pollution permits, or tradable permits, whereby polluting firms are provided with a permit to produce a given level of pollution. This can apply to polluting the air or water. Unlike indirect taxation and regulations, pollution permits are a market based solution as they can be bought and sold. The permits are also subjected to a given level of allowable pollution. The system is referred to as a 'cap and trade' system.

A recent example of a 'cap and trade' system is China's Emission Trading Scheme (ETS). China is the world's largest emitter of greenhouse gases. Much of these

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emissions are carbon dioxide from coal energy production and heavy industry. In 2017, China began implementing pilot carbon markets in seven key economic regions: Beijing, Chongqing, Guangdong, Hubei, Shanghai, Shenzhen and Tianjin. The scheme covers around 25% of China's GDP. The key feature of this innovative scheme is that it controls emissions by providing an incentive to reduce overall emissions. The government 'caps' the total amount of carbon which can be emitted by industry in each region; it then creates a carbon market ('trade') where emitters can buy and sell emission credits. A firm that reduces its emissions will have spare credits which it can then sell to other firms who have exceeded their carbon allowance. China's ETS scheme is being extended to other regions and a wider range of industries. Similar schemes operate in Australia, Singapore, the EU and the USA.

Figure 38.3 shows how the 'cap and trade' system operates. When introduced, the supply of pollution permits is S with a demand of D . The market price is therefore P . Assume over time that demand increases to D_1 (more pollutants are being emitted due to production growth). With supply fixed at S , the price rises to P_1 , giving every incentive to polluting firms to be more environmentally efficient. A more ambitious approach, as being applied in the US, is to reduce the number of permits available over time. Hence, the supply curve shifts left to S_1 , with the price of permits rising to P_2 . This puts even more financial pressure on firms to invest in cleaner technologies, so cutting down on greenhouse gas emissions.

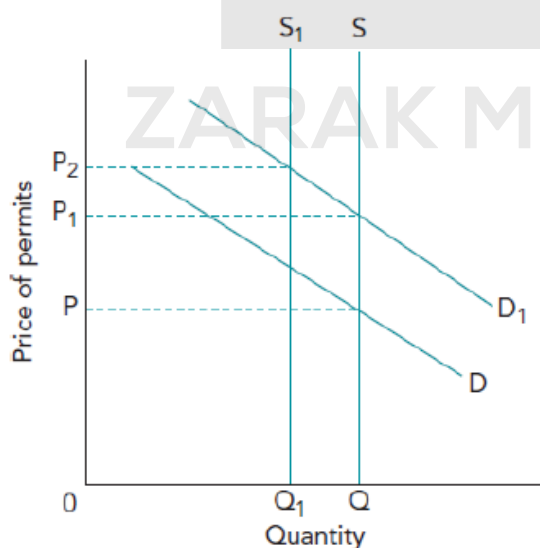


Figure 38.3: Market for tradable pollution permits

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The use of pollution permits has many attractions but they are not without their critics. The European Union's (EU) Emissions Trading System (ETS) has suffered at times from an oversupply of permits, mainly as a consequence of a decline in industrial production coincident with recession in 2008 and 2013. As the EU moved out of recession, the price of permits increased. Low prices for permits are unlikely to provide the incentive many firms need to reduce their greenhouse gas emissions. One likely solution, as indicated in Figure 38.3, is to cut back on the supply of permits.

Price controls, provision of information and production: Another fiscal device is to impose a minimum price on products that generate negative externalities of consumption. This intervention occurs in many countries to limit the consumption of high-sugar energy drinks and tobacco products such as cigarettes. The minimum price increases the price to consumers above the point where $MPB = MSC$. The effectiveness of this policy depends upon the price elasticity of demand for the product. In the case of demerit goods, demand is invariably price inelastic. This will affect the success of a minimum price scheme.

To be more effective, the provision of information can operate alongside fiscal measures to decrease consumption of demerit goods. The provision of information includes warnings and photographs printed on the otherwise plain packets of tobacco products to deter smoking; there is also information provided on the packaging of many processed food products to indicate their sugar and salt content as well as their nutritional qualities. Bottled water usually contains details of the composition of the contents. Another example is for air travel, where it is possible for passengers to know the carbon footprint they are causing and how this might be offset.

Regulations are in place in many countries to restrict the consumption of demerit goods. These regulations usually specify a minimum age limit for consumption. Another way of restricting consumption is through production quotas. This involves limiting the quantity of goods that is produced. If the quota is set at a low level, the price of the product is likely to rise, reducing the amount consumed. Licensing

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suppliers is also used to control supply. One problem of such measures is that they can lead to informal market dealings.

Behavioral insights and 'nudge' theory: The basis of nudge theory lies in the provision of information. It is argued that by presenting choices in a better way, people make better decisions. This avoids the need for more formal government intervention since, when applied, individuals retain their freedom to choose. In some respects, it is a form of paternalism whereby governments or individuals interfere with the affairs of people against their will, yet with their best interests in mind.

Therefore, nudge theory is a way of achieving beneficial economic and social outcomes without the need for regulations. In the case of market failure, it can be applied through social media, letters, emails or personal communications. A typical example is in the case of a free inoculation (merit good) that is available to all people over 60 years old. These people can be targeted by a simple letter that makes them fully aware of the benefits of having the inoculation and how they can get one. In this way, they are 'nudged'; without such a nudge, take up would likely be less and, over time, involve greater costs to the health service.

A second example is in the case of a city that is seeking to reduce the number of cars on its roads (negative externality). There could be a media campaign that focuses on the benefits of cycling or of using other more environmentally acceptable forms of transport like buses. This campaign could nudge motorists into using another form of urban transport.

There is also growing pressure on getting travelers to be more responsible in their use of air transport. Low cost flights seem attractive to customers but invariably they generate much non-essential travel. This travel increases global emissions. A nudge approach would be for a government or even airlines to make these consumers aware of the damage their flights are causing. Alternatives, like rail or the benefits of staying at home, could be highlighted.

In early 2020, in response to the COVID-19 pandemic, many governments produced a regular series of media features advising people what to do to slow down the

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spread of the disease. Sweden, for example, concentrated on 'nudging' people to be responsible to delay the spread of the virus.

The key question is 'Does nudge work?' The answer is that it does, but to only a limited extent. Nudge is best used and most effective when used alongside other policies that are already dealing with market failure.

Causes and consequences of government failure

There are three main causes of government failure: imperfect information, unintended consequences and policy conflict.

Imperfect information: Once the government starts to intervene in the running of markets, it needs information. The correct policies can only be introduced if governments have the correct information. The problem is that governments may have inaccurate or incomplete information. Consequently, they may introduce policies that lead to economic inefficiency. Some examples of this problem are:

- There is a lack of information about the true cost or value of a negative externality. It is often very difficult to give an accurate figure for the cost of a negative externality, such as air pollution. It is also likely to be difficult to trace the source of the pollution. The problem is that it becomes very difficult to impose the correct value of a tax that attempts to reduce production of a polluting firm, for example. The wrong level of tax will lead to the wrong level of production and an even greater inefficiency in the use of resources.
- A similar example is in the case of road pricing charges. There are many economists who advocate road pricing; but there are few real world examples that economists can draw upon to be certain about what the charge should be. If the charge is set too high or too low then there will be a misallocation of resources.
- Governments may lack information about the consumer demand for a merit good such as free healthcare. This is usually because they have no precedent to draw upon. If the government is providing a product free of charge to the consumer, some estimation of the level of consumer demand is required. This is necessary in order to establish the cost of the intervention. This problem also occurs with the provision of public goods. The government must try to

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provide the right amount of such goods. If it does not estimate the level of demand accurately, then the wrong amount will be produced or funded; this is another case of an inefficient allocation of resources.

Unintended consequences

A further problem that arises with government intervention is that there can be unintended consequences. These undesirable intentions can create inefficiencies. Some examples of the ways in which this can happen are:

- The imposition of taxes can distort incentives. High marginal rates of income tax can create disincentives for people to work harder and so gain more income. If this happens, then scarce resources are not being used to their best effect and there is inefficiency. Another similar example is where a government pays benefits to unemployed workers. An unintended consequence is that if the benefits paid are too high, there may be no incentive for the unemployed to look for work.
- Politicians are often seen as being motivated by the desire to remain in government. If this is so, then economic policies may be designed by governments to try to retain power rather than to try to ensure maximum economic efficiency. For example, road pricing is a very unpopular tax in the eyes of car users, despite its tremendous potential to reduce congestion and hence correct market failure. It is likely to be avoided due to a government's fears that its introduction could lead to a loss of electoral votes.

Use of cost and benefits in analyzing decisions

A useful technique to aid decision-making is to analyze the various costs and benefits that are involved. It is particularly useful where a financial appraisal alone may not be entirely appropriate. Analysis of costs and benefits is used by the public sector, for example in the case of public goods, such as a road where no direct charge is made to users. Where the full range of costs and benefits is considered then this involves what is referred to as a cost-benefit analysis.

In general terms, cost-benefit analysis takes a long view and a wide view in assessing whether a particular project should go ahead. The long view comes from the nature of most public sector investment. Because of the large sums involved, a

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project like a new road or rail link is normally appraised over 30 to 50 years. The wide view comes from the need to take into account the full social costs and social benefits. This is because we are often concerned with situations where major economic projects produce substantial and often controversial side effects, in particular where there are externalities that fall upon people and communities who have no direct connection with the particular project.

There are many situations where cost-benefit analysis can be used to aid decision-making. In all types of economy there are numerous examples of environmental pollution that result in external costs being imposed on the local community. The external costs can be far-reaching and substantial. Cost-benefit analysis attempts to quantify the opportunity cost to society of the various possible outcomes or sources of action.

Cost-benefit analysis is also widely used in the appraisal of major transport projects such as the UK's High Speed 2 (HS2) railway and the Hong Kong- Zhuhai- Macau Bridge in the Pearl River Delta region. Where huge investment is needed, cost-benefit analysis can provide the justification to go ahead (or not) based on a broad view of a project's impact on users of the facility and on the wider community and economy over a given period of time.

The cost-benefit approach differs from private sector methods of appraisal in two main ways. First, it seeks to include all of the costs and benefits, not just private ones. Secondly, it often has to provide a shadow price for costs and benefits where no market price is available. For example, in the case of transport projects, there is a need to value travel time savings. Other situations are how to value the benefit of cleaner air or being able to live in a less noisy environment.

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Development of a cost-benefit analysis

Whatever the problem under investigation, there are four main stages in the development of a cost-benefit analysis. The stages are shown in Figure 33.10.

The first stage is to identify all of the relevant costs and benefits arising out of any particular project. This involves establishing what are the private costs, the private benefits, the external costs and the external benefits. This may seem a relatively simple task. In reality, and with a little more thought, it is not so easy. There are particular problems when it comes to identifying external costs and benefits. The costs and benefits are often controversial and may not be easy to define in a discrete way. There may be the added difficulty that it is not always possible to identify a physical or geographical cut-off. The spillover effects of a new road or rail route, for example, are wide-reaching and affect people and communities beyond the immediate location of the proposed development.

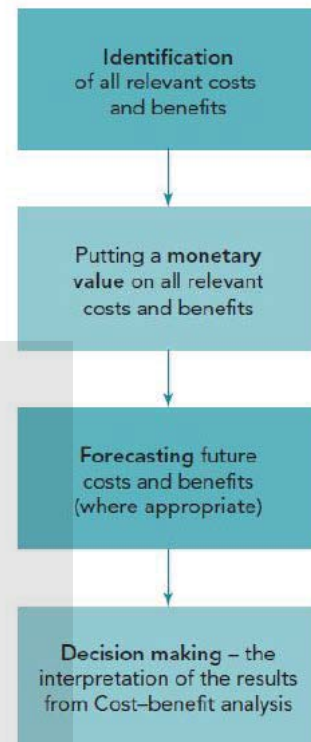


Figure 33.10: Stages of a cost-benefit analysis

The second stage involves putting a monetary value on the various costs and benefits. This is relatively straightforward where market prices are available. For example, in the case of a new road, a monetary value can be put on the jobs created or on the supplies provided by local businesses. For other variables, where no direct prices are available, a monetary value must be attributed for the costs and benefits. This is difficult and is controversial where cost-benefit analysis has come under close public scrutiny. A good example is how to value time, especially savings from a faster travel time.

The third stage applies in situations where projects have longer-term implications that extend well into the future. Here, economists have to employ statistical

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forecasting techniques, sometimes of a very crude nature, to estimate costs and benefits over many years. This particularly applies to projects where massive capital expenditure is involved.

The final stage is where the results of the earlier stages are drawn together so that the outcome can be presented in a clear manner in order to aid decision-making. The important principle here is that if the value of benefits exceeds the value of the costs, then the particular project is worthwhile since it provides an overall net benefit to the community. An estimate is made of the benefit: cost ratio that is the net benefits as a proportion of the net costs of a project. Dependent on funding, projects with the highest benefit: cost ratios are most likely to get the go ahead.

The four stages in a cost-benefit analysis provide a clear framework for making decisions where the market mechanism is not fully functional. Any application of the technique should be seen in this way.

The advantages and disadvantages of cost-benefit analysis

The advantages of cost-benefit

- It takes into account all of the various costs and benefits resulting from a proposed investment project.
- Many of the various costs and benefits will have market prices attached to them, making it relatively easy to calculate

The disadvantages of cost-benefit

- It may be difficult to identify all of the relevant external costs and external benefits. For example, how wide will the spill-over or third-party effects be? It may therefore not be easy to decide on which costs and benefits to include in the analysis.
- Some of the costs and benefits will not have market prices attached to them and this will make it much more difficult to calculate their monetary value.

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monetary values.

To overcome this problem, shadow prices can be used to estimate such values, but these shadow prices may be very difficult to estimate.

- The process can analyze costs and benefits today, but also well into the future. This has the advantage that the cost-benefit analysis can take a long form view of the possible consequences of an investment project.
- The estimation of costs and benefits well into the future can be very difficult. There are forecasting techniques which can be used, but it is very difficult to estimate the costs and benefits of a project 10 or 15 years into the future. This raises the issue of the time value of money, i.e. where the value of the costs and benefits today will be different from their value in 10 or 15 years' time; this will therefore involve a process of discounting these future values.
- All of the information obtained as a result of the cost benefit analysis can be used to ensure that the correct decision is taken it is therefore a very significant aid to the decision-making process.
- Cost benefit analysis is usually used to assess investment projects in the public sector and so even if the analysis suggests that an investment project would yield a net benefit to a community, there may be political reasons why it does not actually go ahead.

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